

Experiment No. 30

Title

Prolog Program for Backward Chaining

Aim

To write a Prolog program to implement **Backward Chaining**.

Objective

To understand how a goal is proved by working backward through rules and known facts.

Algorithm

1. Start.
2. Define facts and rules.
3. Set the required goal.
4. Find a rule that can prove the goal.
5. Check whether its conditions are satisfied.
6. If satisfied, prove the goal.
7. Display the result.

Flowchart

Start

|

v

Define Facts & Rules

|

v

Set Goal

|

v

Find Matching Rule

|

v

Check Conditions

|

v

Conditions True?

/	\
Yes	No
v	v

Prove Goal Backtrack

v

Display Result

v

Stop

Prolog Program

```
fact(rain).
```

```
rule(rain, wet_ground).
```

```
backward_chain(Goal) :-
```

```
    fact(Fact),
```

```
    rule(Fact, Goal).
```

Query

```
?- backward_chain(wet_ground).
```

Sample Output

```
true.
```

Result

The Prolog program successfully implemented **Backward Chaining** and proved the required goal from the available facts and rules.

Conclusion

Thus, Backward Chaining was successfully demonstrated in Prolog by starting from the goal and checking the supporting facts.